

When Emotion Slows You Down: Valence-Arousal Conflicts Across L1 & L2

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We tested whether the Valence × Arousal interaction differed across:
tasks (Arousal judgement vs. Valence judgement), **language nativeness** (L1 vs L2) and **stimulus type** (words vs. pictures)
Valence and Arousal interacted to influence responses across three different stimulus types: **Pictures, L1 words and L2 words**.
Task demands had the largest effect on word responses: for word stimuli, the Valence-Arousal Conflict Pattern was only found in the **arousal judgement task**
There were **no differences in L1 vs L2 processing** for highly proficient German-English bilinguals: No Foreign Language Effect

BACKGROUND

Dimensional Theories of Emotion:

Valence = is stimulus **positive** or **negative**
Negative valence → withdrawal tendency
Positive valence → approach tendency

Arousal = is stimulus **calming** or **exciting**
High arousal → withdrawal tendency
Low arousal → approach tendency

Valence-Arousal Conflict Theory:
valence X arousal are interactive, not cumulative:

High arousal + negative = congruent
(avoid + avoid) = **faster response times**

Low arousal + positive = congruent
(approach + approach) = **faster response times**

However:

It is unclear whether this affects word stimuli the same as picture stimuli ?
It is unclear whether this effect can be found in L2 for bilingual speakers ?



RQ1: stimulus type

Does the Valence x Arousal interaction differ across stimulus type when comparing pictures vs. words?

RQ2: language

Does the Valence x Arousal interaction differ across L1 and L2 words (e.g., nativeness) for bilingual speakers?

RQ2: task demands

Does the Valence x Arousal interaction differ across task type and the affective dimension of classification (valence or arousal)?

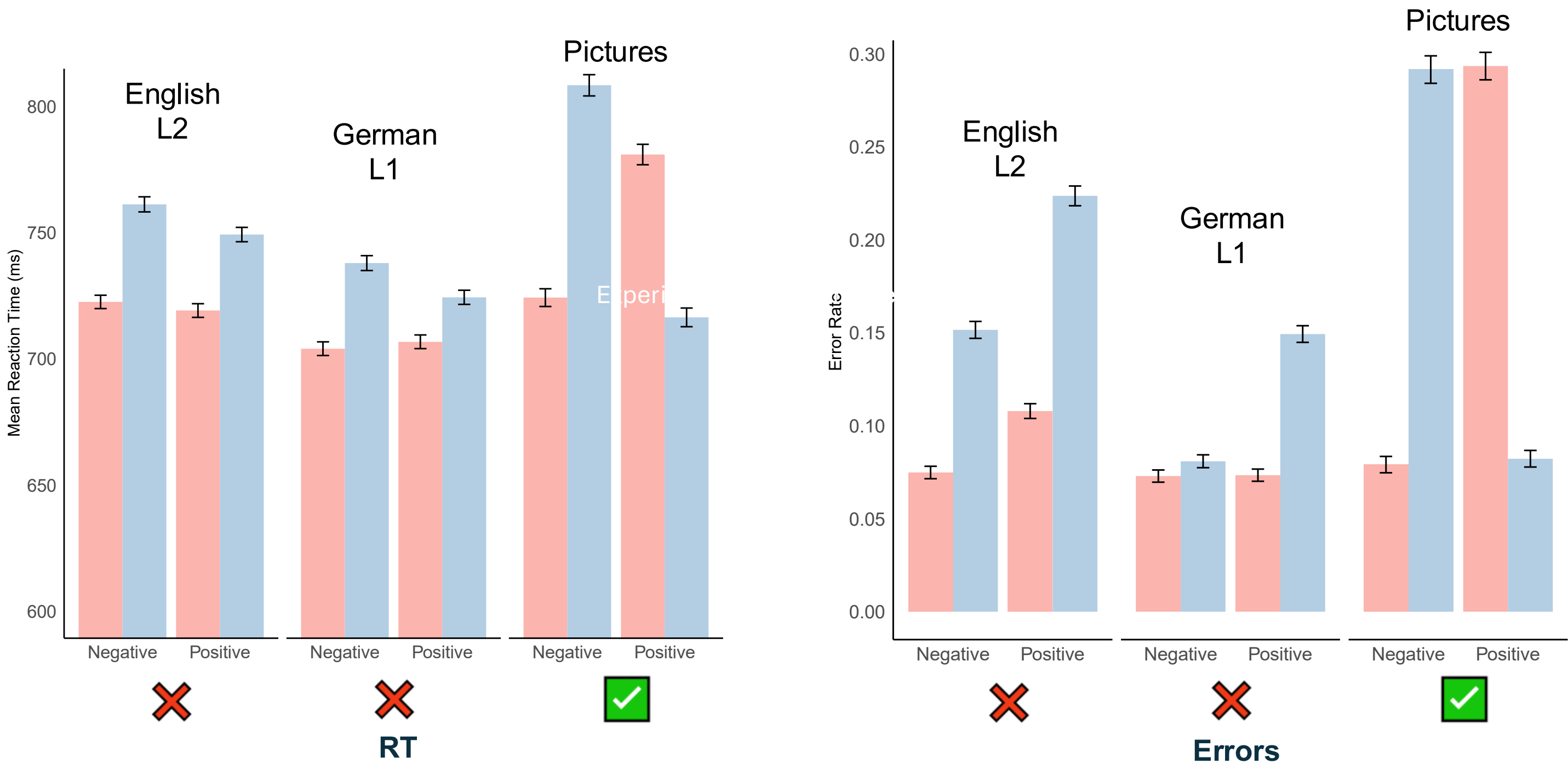
ANALYSIS

RT ~ Valence × Arousal × Stimulus Type + Age + Education + (1 + Stimulus Type | Participant) + (1 | Item)
- Accuracy was analysed using logistic mixed-effects models with the same fixed-effect structure.

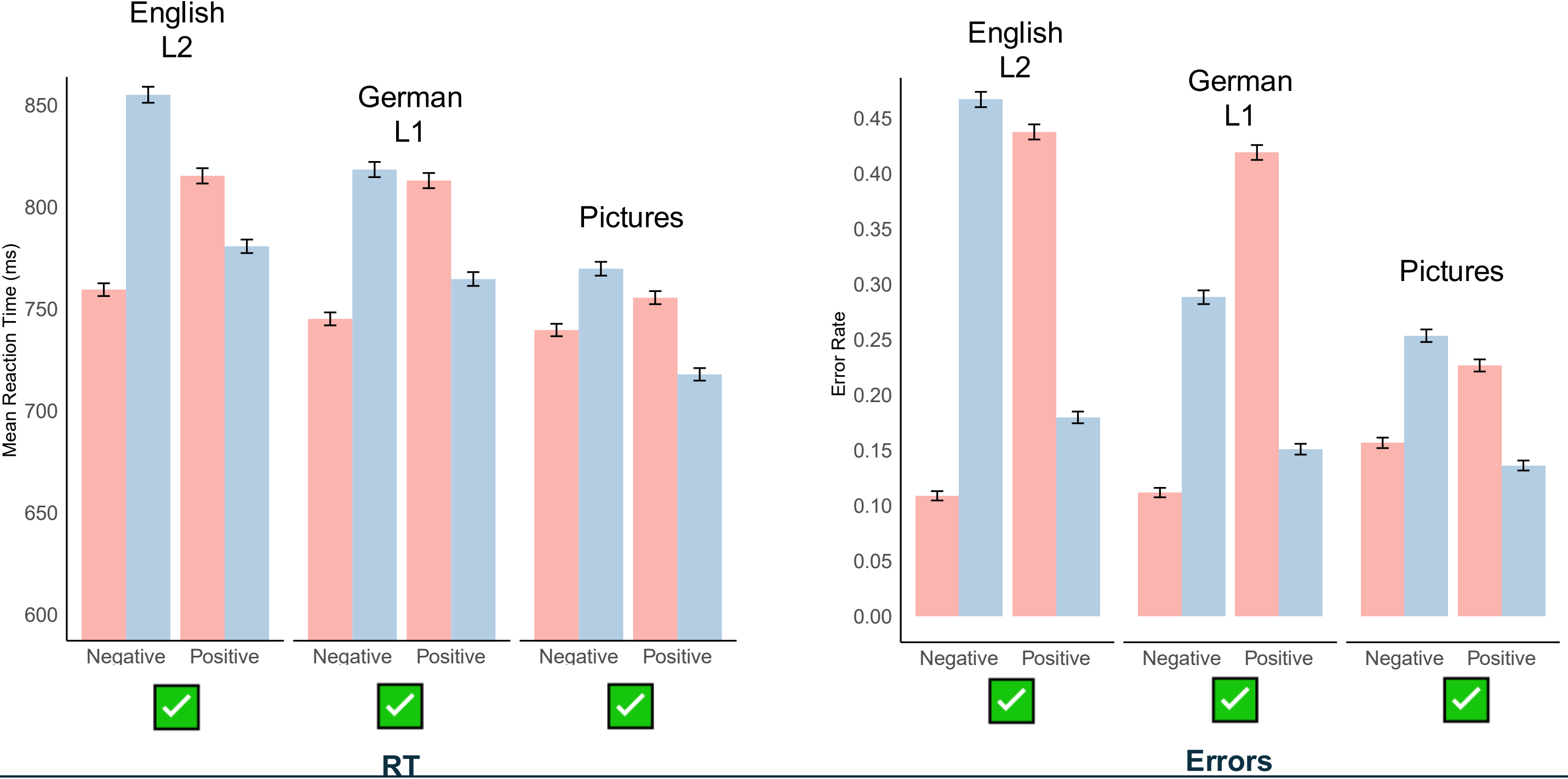
Analyses: (1) main model, (2) L2 model (with moderators) for both exps
Exclusions: RTs <200 ms or >2500 ms, categorisation errors, outliers beyond ±3 SD, comprehension check fails

RESULTS

EXPERIMENT 1: Valence Judgement



EXPERIMENT 2: Arousal Judgement



CONCLUSION: Valence X arousal interacted → affective dimensions should not be analysed independently.

-Valence and arousal interacted to shape response times in both **EXP1** ($b = -65.47, p < .001$) and **EXP2** ($b = -114.81, p < .001$),
-For words, the pattern emerged **only during arousal judgements**, however it was **robust for picture stimuli**
-There was **no evidence that Valence x Arousal interaction differed in L1 and L2 words** in either **EXP1** ($b = 0.55, p = .989$) or **EXP2** ($b = 0.55, p = .662$).

However: In EXP 1, **Earlier Age of Acquisition** predicted faster responses to **low-arousal words** ($b = 2.39, p = .018$). In EXP2, **Higher Daily L2 Use** predicted slower responses to **high-arousal words** ($b = -0.59, p = .005$), **Higher L2 Family Immersion** was associated with slower responses to **negative versus positive words** ($b = -1.40, p = .017$) AND **low-versus high-arousal words** ($b = 1.28, p = .031$).

METHOD

Participants

Two independent samples of German-English sequential bilinguals living in Germany completed the study:

	Experiment 1	Experiment 2
	Valence task	Arousal task
<i>n</i>	158	133
Lexical Score %	85.37 (11.68)	85.19 (11.72)
L2 Reading Proficiency (11-pt)	9.45 (1.17)	9.54 (0.99)
L2 Age of Acquisition	8.94 (2.65)	9.10 (3.05)
L2 Daily Use %	33.49 (21.26)	34.08 (22.46)
L2 family immersion (years)	4.48 (8.48)	2.42 (7.53)

Participants were recruited through Prolific and German universities. All had German as their L1 and were similar across age, education, and bilingual demographics.

Materials

Word stimuli: 40 German-English translation-equivalent word pairs matched pairwise on: valence, arousal, frequency, length

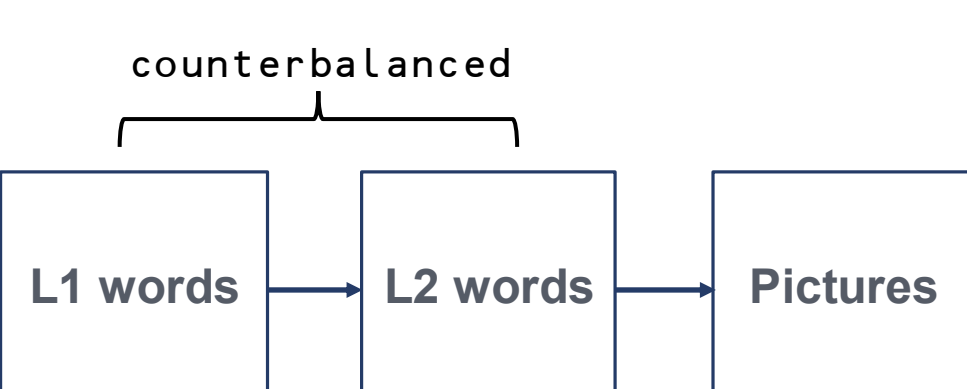
English words selected from ANEW
German words selected from ANGST

Picture stimuli: 48 images selected from the Nencki Affective Picture System (NAPS)
12 in each Valence × Arousal condition (48 total)

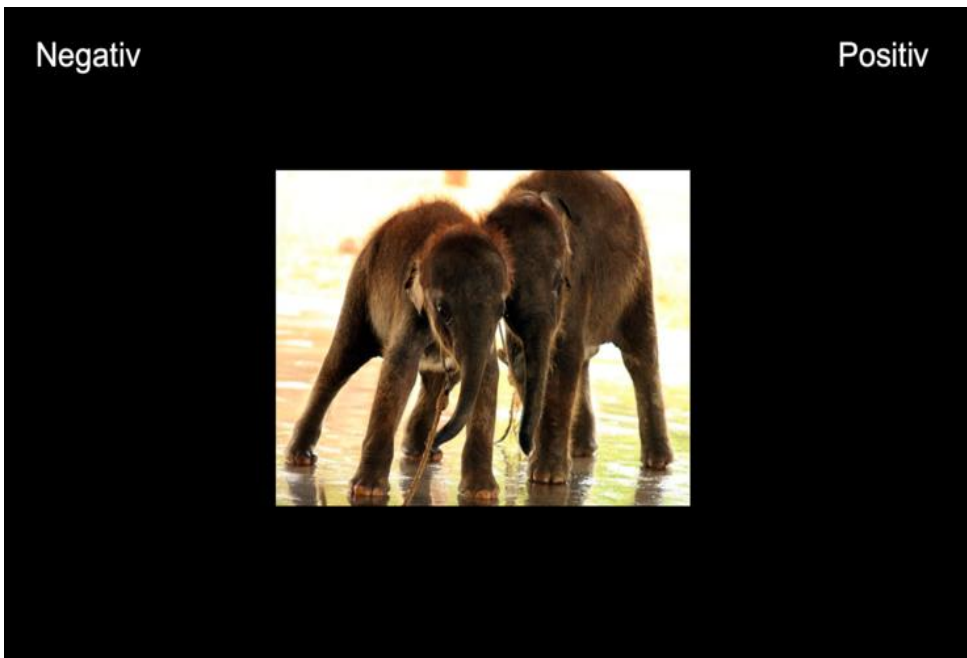
Pictures followed the same 2 × 2 affective structure as the word stimuli

The same stimulus sets was used in both experiments

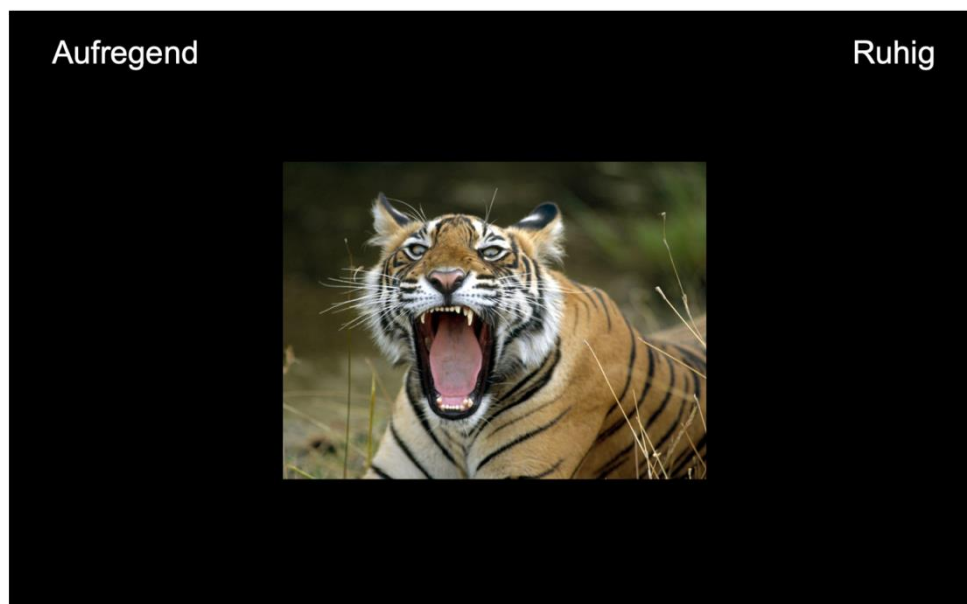
Procedure



Experiment 1: positive versus negative



Experiment 2: calming versus arousing



2 Valence × 2 Arousal × 3 Stimulus Type

- 40 words - presented four times.
- 160 trials per language
- 48 pictures - presented twice.
- 96 picture trials
- 416 experimental trials per participant

Materials, stimuli and additional analyses:



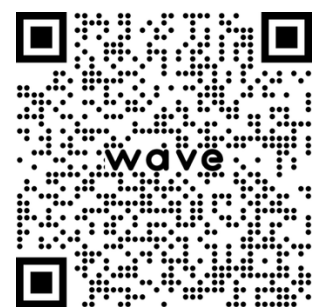
Positive/pleasant			
Low arousal		High arousal	
Straße	street	bewundert	admired
Fenster	window	Jubel	joy
Baum	tree	Leidenschaft	passion
ernähren	nourish	Geld	money
sauber	tidy	strahlend	radiant

Negative/unpleasant			
Low arousal		High arousal	
unförmig	deformed	Spinne	spider
dunkel	dark	Angriff	assault
Harn	urine	demütigen	humiliate
trüb	muddy	Geisel	hostage
starr	rigid	Kakerlake	cockroach

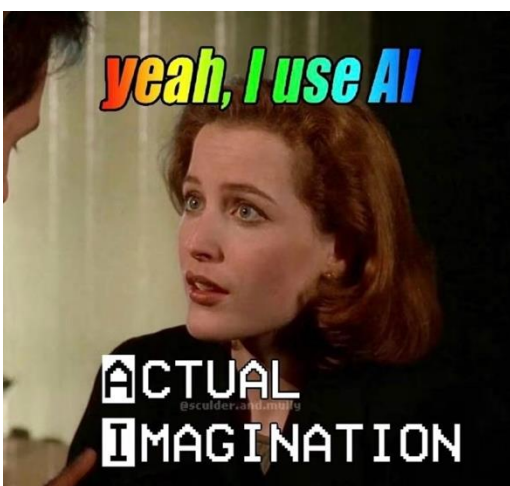
Declarations:

Thank you for your time!

CONTACT ME:



PROJECT SITE:



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Ethics: Approved by Curtin University Human Research Ethics Committee, HREC2020-0026.

